

MTH313M End Semester Examination (May 4, 2026) Part B

5:00 pm - 7:15 pm (+45 minutes for DAP students) Maximum Marks: 15

Instructions:

- (a) Justify your answer in all the questions attempted.
- (b) Please start the answer of a new question on a new page of your answer-script.

- Question 1. (3 + 2 marks) Let $\{X_t\}_{t \geq 0}$ denote the number of customers in a $M/G/1$ queueing system with arrival rate $\lambda > 0$. Assume that the service time distribution is given by a probability density function f and that $S_1, S_2, \dots$ denote the departure times of the customers after completing their service. Set $S_0 = 0$. Show that $\{X_{S_n} : n = 0, 1, 2, \dots\}$ forms a (discrete time) Markov chain and find its transition probability matrix.
- Question 2. (5 marks) Let $\{X_t\}_{t \geq 0}$ denotes a Yule process with initial population 2 and infinitesimal birth parameter $\lambda > 0$. Find the probability mass function of X_t for all $t \geq 0$.
- Question 3. (4 + 1 marks) Let $S_1, S_2, \dots$ be the arrival times of the events observed in a Poisson Process $\{N_t\}_{t \geq 0}$ with rate $\lambda > 0$. Fix $t > 0$ and let $k \geq 2$ be a natural number. Find the conditional joint p.d.f. $f_{S_1, S_2, \dots, S_k | N_t = k}$ and hence find the conditional joint distribution function $F_{S_1, S_2, \dots, S_k | N_t = k}$.